

DATA STRUCTURES PROJECT REVIEW TEMPLATE (50% COMPLETION)

Project Mid Review Evaluation Sheet

1. Student Details

Project Number	24
Project Title	PetCare – Smart Veterinary Clinic & Pet Health Management System
Student Names	Sai Nikhil Vukka G.V Sai Manikanta R.S.V.D.S Mohitswar
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Branch / Section	
Guide Name	
Review Date	28-05-2026

2. Project Slides

Abstract	PetCare is a Java-based command-line application that models a veterinary clinic and pet health management platform. It manages pet profiles, medical histories, veterinary appointments, vaccination schedules, and clinic analytics using six categories of classical DSA — Trees, Multiway Trees, Graphs, Shortest Paths, Sorting, and Greedy/DP — applied to realistic clinic operations.
Problem Statement	Traditional veterinary management systems lack efficient data-structure-backed operations for indexing pet records, optimising routes between clinic branches, and scheduling overlapping appointments. PetCare bridges this gap by mapping each DSA module to a concrete clinical problem domain, demonstrating practical application of all six course modules within a single cohesive system.
Objectives of Project	1. Implement BST & AVL Trees for $O(\log n)$ pet record indexing. 2. Use B-Tree, B+ Tree, Segment Tree & Fenwick Tree for multiway indexing and range queries on vaccination schedules. 3. Apply BFS, DFS, and MST (Kruskal/Prim) for clinic network management. 4. Implement Dijkstra, Bellman-Ford, Floyd-Warshall, and Topological Sort for route and procedure scheduling. 5. Use Merge, Quick, Heap, Radix, and Counting Sort for treatment record ranking. 6. Apply Activity Selection, Fractional & 0/1 Knapsack, and LIS for scheduling and resource allocation.

Domain / Application Design	Domain: Veterinary Clinic & Pet Health Management Architecture: Java CLI app with nested menus — one sub-menu per DSA module. Entities: Pet, Treatment, Appointment, Vaccination, Clinic Packages: petcare.models, petcare.utils, petcare.menu, petcare.trees, petcare.graphs, petcare.sorting, petcare.algorithms
Selected Data Structures	BST, AVL Tree, B-Tree (order t), B+ Tree, Segment Tree, Fenwick Tree, Graph (Adjacency List), Priority Queue, Arrays
Programming Language Used	Java (JDK 11+)
Software / Tools Used	Eclipse / IntelliJ IDEA / VS Code, Java JDK, Command-Line Interface (CLI), Git
Sample Input/Output Shown	Yes — CLI nested menus demonstrate each algorithm on sample pet and clinic data (e.g., AVL insert/search on pet IDs, Dijkstra on clinic graph, Merge Sort on treatment costs, Activity Selection on appointments).

3. Demonstration Checklist

Criteria	Status
Problem Statement Clearly Explained	Yes
Working Prototype Available	Yes
Data Structures Properly Used	Yes
Code Execution Demonstrated	Yes
Sample Input/Output Shown	Yes
Database Connectivity (if any)	No
Error Handling Implemented	Yes
UI / Interface Developed	No (CLI)

4. Evaluation Rubrics

Parameter	Max Marks
Problem Understanding	10
Use of Data Structures	15
Algorithm Design	15
Coding Progress	15
Innovation / Creativity	10
Demonstration	10
Documentation	10
Presentation Skills	10

Total	100
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